

CS683 Project Assignment

iperf3 network tester
Jerold (Jerry) Abramson

1. Overview	3
1.1. More Details	3
2. Video Presentation	3
3. Functional Requirements and limitations	3
3.1. Challenges resolved	4
3.2. Future Goals	4
4. Motivation	4
5. Purpose	4
5.1. Potential Users	4
5.2. Related Work	5
5.2.1. Text-Based Mode	5
5.2.2. “Fancy” line drawing and fonts	5
5.2.3. Similar Android Applications	6
5.2.4. Similar iOS application	6
6. Rall Requirements have been completed	6
6.1. Requirement 1	6
6.2. Requirement 2 – User Interface implemented	7
6.3. Requirement 3	7
6.4. Requirement 4 – Requirement removed due to further application analysis.	7
7. Design and Implementation	8
7.1. Layered Architecture	8
7.2. Test Execution Flow	8
7.3. Full component breakdown organized by prose	8
2. Project Structure	9
8.	9
9. Screen shots	10

9.1.	Real Android Phone	10
9.2.	Android Emulator	13
	Timeline	26
10.		26
11.	Current Status - Complete	27
11.1.	Fallback simulation	27
12.	AI Usage Log	28
13.	Future Work (Optional)	28
•	Project Demo Links	28

1. Overview

The application being developed will utilize the de-facto network protocol tool, **iperf3**, for testing network speeds between a client and a server.

1.1. More Details

- For my project I am implementing a network performance monitor based on the iPerf3 protocol.
- Although there are a couple of Android apps that already partially implement this, one of them is terribly complex, quite expensive, (**Analti**), more focused on Wi-Fi and internet performance monitoring.
- There is also a very old version that is nothing more than the text-based version of the existing 'iperf3' application that runs in a console terminal window.
- There is a nice version that runs on iOS, but that has not been ported to Android. However, the iOS version sacrifices details for ease of use and a clean user interface.
- Here are the associated application icons:

My Project	Analti by analiti Networking Experts	Iperf3 by Uncle Tools
TBD	https://play.google.com/store/apps/details?id=com.analiti.fastest.android	https://play.google.com/store/apps/details?id=tools.uncle.network
		

2. Video Presentation

I broke up the video presentation into 4 videos, roughly covering different aspects of the deliverables.

The videos are unlisted public links:

Video Description	YouTube link
CS683 - Project - Part 1 of 4 - Jerold Abramson	https://youtu.be/duv60JoqVTs
CS683 - Project - Part 2 of 4 - Jerold Abramson	https://youtu.be/5vFkzoGIDoU
CS683 - Project - Part 3 of 4 - Jerold Abramson	https://youtu.be/2TtQ8dWUnhg
CS683 - Project - Part 4 of 4 - Jerold Abramson	https://youtu.be/iXRrtdUoofw

3. Functional Requirements and limitations

- The application **will only be implementing** the client-side functionality of iPerf3:
 - It will assumed that an **iPerf3** server is running on another host.
 - I will be providing a mechanism to ensure that a server will be available to the instruction team during analysis and grading.
- The actual **iperf3** protocol **will not be** reimplemented.

3.1.Challenges resolved

- I have solved working through the mechanism to ensure that the `iperf3` binary is uploaded to the Android device/emulator.
- I have also solved having the `iperf3` binary being executable inside the standard Android sandbox security architecture.
- I have identified the core issue with executing native executables on Android which is specific to SELinux.
- I have identified a reasonable resolution for this for the project assignment.
- I am using the standard Kotlin/Java ProcessBuilder class APIs.
- I have a script that can be run by the testing team to allow for the `iperf3` executable to run.
- The basics of the workaround are to disable SELinux on the emulator, which allows for the binaries included with the application to be executable.

3.2.Future Goals

- I have studied a GitHub project that incorporates the `iperf3` binary into a full Android NDK application.
- I even moved on further and started converting my application to utilize this approach.
- The native `iperf3` native libraries (`.so`) **will not be** utilized using direct JNI APIs from Kotlin/Android.
- The official releases of the `iperf3` binary for Android (all hardware architectures and emulators) will be utilized for this project.
- I have put that effort into another branch:
 - <https://github.com/CS683/project-jerryabramson/tree/tryembeddediperf3>
- For the course project I will continue with the current approach of using an external `iperf3` binary.

4. Motivation

I utilize network testing tools as part of my home-labb'ing hobby.

One of the tools I am constantly utilizing is based on the `iPerf3` protocol, which is defined here:

The full `iperf3` source code and technical documentation for `iperf3` is available on this website:

- <https://iperf.fr/>

The full source code and technical documentation for the `iperf3` implementation on Android is available here:

- <https://github.com/davidBar-On/android-iperf3/>

I utilize network testing tools as part of my "home-labb'ing" hobby.

- One of the tools I am constantly utilizing is based on the `iPerf3` protocol, which is defined here:
- *For some context on `iPerf3` is available → here*

5. Purpose

Provide the ability to test WiFi and generalized network performance using an Android phone.

5.1.Potential Users

- `iperf3` is a well-known tool for networking professionals.
- My goal is to launch this application on the Android Play Store over the summer for a one-time purchase price of `$0.99`

5.2.Related Work

I have developed a **Java text-based console program** that runs the **iperf3** native executable in the background and provides performance details in an asynchronous manner.

The application supports running at the command-line with ‘fancy’ ANSI line drawing and fonts, as well as standard text-based output suitable for scripts.

Here are some examples of this application running in standard text-based mode

- Note: **pvc** is a local hostname reachable in my LAN.

5.2.1. Text-Based Mode

```
TERM=dumb LANG=C newTestBandWidth.sh pvc -R
[Sat Mar 28 13:10:52 EDT 2026] Executing Command-line: '/opt/local/bin/iperf3' '--forceflush' '--connect-timeout' '3000' '-c' 'pvc' '-O 2' '-P 8' '-t' '10' '-R'
Initiating [ .....Connecting to host pvc, port 5201
Reverse mode, remote host pvc is sending
  Local Host/IP: 192.168.1.10 remote Host/IP: 192.168.1.7 Remote Port: 5201
...XXXXX Skipping 0.00-1.01          2.35 Gbits/sec (omitted)
...XXXXX Skipping 1.01-2.00          2.35 Gbits/sec (omitted)
...XXXXX Running 0.00-1.01           2.35 Gbits/sec
...XXXXX Running 1.01-2.01           2.35 Gbits/sec
...XXXXX Running 2.01-3.00           2.35 Gbits/sec
...XXXXX Running 3.00-4.01           2.35 Gbits/sec
...XXXXX Running 4.01-5.01           2.35 Gbits/sec
...XXXXX Running 5.01-6.00           2.35 Gbits/sec
...XXXXX Running 6.00-7.01           2.35 Gbits/sec
...XXXXX Running 7.01-8.00           2.35 Gbits/sec
...XXXXX Running 8.00-9.00           2.35 Gbits/sec
Running 9.00-10.01          2.35 Gbits/sec
Results: 0.00-10.01          2.35 Gbits/sec sender
=> 0.00-10.01          2.35 Gbits/sec receiver

[Avg]      2.35 Gbits/sec
[Min]      2.35 Gbits/sec
[Max]      2.35 Gbits/sec

-----
Return Code: 000 [Avg=2.35 Gbits/sec]
```

5.2.2. “Fancy” line drawing and fonts

```
newTestBandWidth fedoranas -R
[Sat Mar 28 13:16:38 EDT 2026] Executing Command-line: 'C:/Program Files/iperf3.12_64/iperf3.exe' '--forceflush' '--connect-timeout' '3000' '-c' 'fedoranas' '-O 2' '-P 8' '-t' '10' '-R'
Connecting to host fedoranas, port 5201
Reverse mode, remote host fedoranas is sending
  Local Host/IP: 192.168.1.20 remote Host/IP: 192.168.1.204 Remote Port: 5201
Running 1.00-2.00 - ( [ 7.61 Gbits/sec

newTestBandWidth fedoranas -R
[Sat Mar 28 13:16:38 EDT 2026] Executing Command-line: 'C:/Program Files/iperf3.12_64/iperf3.exe' '--forceflush' '--connect-timeout' '3000' '-c' 'fedoranas' '-O 2' '-P 8' '-t' '10' '-R'
Connecting to host fedoranas, port 5201
Reverse mode, remote host fedoranas is sending
  Local Host/IP: 192.168.1.20 remote Host/IP: 192.168.1.204 Remote Port: 5201
Results: 0.00-10.00 [ 7.31 Gbits/sec sender
=> 0.00-10.00 [ 7.31 Gbits/sec receiver

[Avg] [ 7.31 Gbits/sec
[Min] [ 5.59 Gbits/sec
[Max] [ 9.23 Gbits/sec

Return Code: 000 [Avg=7.31 Gbits/sec]
```

There are a few similar implementations of this tool on the Android play store, shown in the below table:

5.2.4. Similar iOS application

6. Rall Requirements have been completed

The status'es have all been updated to reflect which iteration the requirement was completed in.

Title	prototype of remote IP connection (ESSENTIAL)
Description	A hard-coded remote IP address will be used as the remote iPerf3 server Hard-coded command-line arguments will be used for iPerf3
Acceptance Tests	runs a fixed loop of 10 iperf3 intervals over TCP/IP in the background
Test Results	Initially the output may only be visible via logcat I managed to get the output to display after the test is completed. I still need to work on the asynchronous behavior so we can see the speed results as they are presented (every second).
Mockup	<pre> e du...project_iperf3_network_tester V ANGLE allowlist from config; con.dreamweaves.royalclutch.con.dts.freefirecon.con.dex.firmon.con.gr e du...project_iperf3_network_tester V du.bu.cash83.sjrahman_project_iperf3_network_tester is not listed in ANGLE allowlist or settings. e du...project_iperf3_network_tester V Neither updatable production driver nor pre-release driver is supported. e du...project_iperf3_network_tester V Chromographer implicitly registered for the refresh rate. e du...project_iperf3_network_tester V HostManager\$DeviceInfo(7215976) LocalList changing from [] to [en-US] e du...project_iperf3_network_tester E Pining is deprecated since Android Q. Please use trim or other methods. e du...project_iperf3_network_tester D Pinset change id reported: 37766616; ID: 10224; state: ENABLED e du...project_iperf3_network_tester D Toggle override initialized to OVERIDE_DISABLED e du...project_iperf3_network_tester I Opening libEGLv1_2M emulation.so e du...project_iperf3_network_tester I Opening libGLESv1_2M emulation.so e du...project_iperf3_network_tester W OpenGL ES 3.0-3.1 not supported e du...project_iperf3_network_tester E Failed to choose config with EGL_SWAP_BEHAVIOR_PRESERVED, retrying without ... e du...project_iperf3_network_tester E Failed to initialize EGL12-2 format, error = EGL_SUCCESS e du...project_iperf3_network_tester W Unknown category 0 e du...project_iperf3_network_tester I hideNapi1: Accessing hidden method (android/os/SystemProperties::addChangeCallback(Ljava/lang/Ru e du...project_iperf3_network_tester I Compiler allocated 511348 to compile vite android.view.ViewImpl.performAvarants() e du...project_iperf3_network_tester I hide(Napi1, FrameHandler) e du...project_iperf3_network_tester I edu.bu.cash83.sjrahman_project_iperf3_network_tester.TCPDataView.onCancelled at PHASE_CLIENT_ALREADY e du...project_iperf3_network_tester e du...project_iperf3_network_tester e du...project_iperf3_network_tester I iperf3 binary: /bin/iperf3 e du...project_iperf3_network_tester O stdout: Connecting to host 192.168.1.28, port 5201 e du...project_iperf3_network_tester O stdout: Reverse mode, remote host 192.168.1.28 is sending e du...project_iperf3_network_tester O stdout: [*] Local 10.8.0.2.1a port 42036 connected to 192.168.1.28 port 5201 e du...project_iperf3_network_tester O stdout: [0] Interval Transfer Bitrate e du...project_iperf3_network_tester O stdout: [1] 0.00-0.00 sec 42.0 Mbytes 392 Mbits/sec e du...project_iperf3_network_tester O stdout: [1] 1.00-2.00 sec 48.1 Mbytes 403 Mbits/sec e du...project_iperf3_network_tester O stdout: [1] 2.00-3.00 sec 48.2 Mbytes 403 Mbits/sec e du...project_iperf3_network_tester O stdout: [1] 3.00-4.00 sec 47.6 Mbytes 399 Mbits/sec e du...project_iperf3_network_tester O stdout: [1] 4.00-5.00 sec 48.2 Mbytes 404 Mbits/sec e du...project_iperf3_network_tester O stdout: [1] 5.00-6.00 sec 47.2 Mbytes 396 Mbits/sec e du...project_iperf3_network_tester O stdout: [1] 6.00-7.00 sec 48.3 Mbytes 404 Mbits/sec e du...project_iperf3_network_tester O stdout: [1] 7.00-8.00 sec 48.5 Mbytes 407 Mbits/sec e du...project_iperf3_network_tester O stdout: [1] 8.00-9.00 sec 48.8 Mbytes 415 Mbits/sec e du...project_iperf3_network_tester O stdout: [1] 9.00-10.00 sec 49.2 Mbytes 413 Mbits/sec e du...project_iperf3_network_tester O stdout: - - - - - e du...project_iperf3_network_tester O stdout: [0] Interval Transfer Bitrate Retr sender receiver e du...project_iperf3_network_tester O stdout: [1] 0.00-10.00 sec 482 Mbytes 404 Mbits/sec 0 0 e du...project_iperf3_network_tester O stdout: [1] 0.00-10.00 sec 482 Mbytes 404 Mbits/sec 0 0 e du...project_iperf3_network_tester O stdout: iperf Done e du...project_iperf3_network_tester I iperf3 binary: /bin/iperf3 e du...project_iperf3_network_tester O stdout: Connecting to host 192.168.1.28, port 5201 e du...project_iperf3_network_tester O stdout: Reverse mode, remote host 192.168.1.28 is sending e du...project_iperf3_network_tester O stdout: [*] Local 10.8.0.2.1a port 42036 connected to 192.168.1.28 port 5201 e du...project_iperf3_network_tester O stdout: [0] Interval Transfer Bitrate e du...project_iperf3_network_tester O stdout: [1] 0.00-1.00 sec 42.0 Mbytes 352 Mbits/sec e du...project_iperf3_network_tester O stdout: [1] 1.00-2.00 sec 46.3 Mbytes 392 Mbits/sec e du...project_iperf3_network_tester O stdout: [1] 2.00-3.00 sec 48.8 Mbytes 403 Mbits/sec e du...project_iperf3_network_tester O stdout: [1] 3.00-4.00 sec 47.6 Mbytes 399 Mbits/sec e du...project_iperf3_network_tester O stdout: [1] 4.00-5.00 sec 48.2 Mbytes 404 Mbits/sec e du...project_iperf3_network_tester O stdout: [1] 5.00-6.00 sec 47.2 Mbytes 387 Mbits/sec e du...project_iperf3_network_tester O stdout: [1] 6.00-7.00 sec 48.3 Mbytes 397 Mbits/sec e du...project_iperf3_network_tester O stdout: [1] 7.00-8.00 sec 48.5 Mbytes 389 Mbits/sec e du...project_iperf3_network_tester O stdout: [1] 8.00-9.00 sec 46.7 Mbytes 392 Mbits/sec e du...project_iperf3_network_tester O stdout: [1] 9.00-10.00 sec 43.6 Mbytes 366 Mbits/sec e du...project_iperf3_network_tester O stdout: - - - - - e du...project_iperf3_network_tester O stdout: [0] Interval Transfer Bitrate Retr sender receiver e du...project_iperf3_network_tester O stdout: [1] 0.00-10.00 sec 483 Mbytes 388 Mbits/sec 0 0 e du...project_iperf3_network_tester O stdout: [1] 0.00-10.00 sec 482 Mbytes 397 Mbits/sec 0 0 e du...project_iperf3_network_tester O stdout: iperf Done </pre>
Status	Iteration 1 - Completed

6.2.Requirement 2 – User Interface implemented

Title	User Interface Displays Interaction and Status
Description	After the user presses the button ‘Run Iperf-3 Test’ the execution runs in the background, with the user interface providing a progress bar.
Mockups	
Acceptance Tests	The progress bar moves forward while the background job is running.
Test Results	The progress bar moves forward, and the ‘Run Iperf-3 Test’ button is greyed out. Once the background job is completed, the progress bar shows ‘Complete’ and the button is again usable (no longer greyed out)
Status	Iteration 1 – Job runs in background Iteration 2 – As the job is running in the background, the progress bar is updated with the most recent bandwidth result i.e [=====] Interval 5: 393 Mbits/sec Iteration 2 – Accurate results based on the optional inputs (i.e. hostname, run count, parallel streams, etc...) Iteration 3 – All typical iperf3 options provided (duration, parallel streams, upload/download, omitted tests)

6.3.Requirement 3

Title	Server Address and other options
Description	The user shall be able to type in a valid server address of a remote iperf3 server
Mockups	
Acceptance Tests	host-name/IP Address validation
Test Results	Positive Test: "localhost" Test runs Negative Test: "abcd.baddomain" <i>Validation error</i>
Status	Iteration 1 - Allow entry of server name Iteration 2 - Validates remote host before execution Iteration 3 - perform iperf3 test with hard-coded-count of 10 against provided host Iteration 4 – All command-line arguments supported: • Server Address • Server Port • Transmit Mode (Upload/Download) • Parallel Streams • Skip Iteration Count • Test Duration • Output Units – No longer makes sense for a mobile solution. • Loop count – Will be implemented for play store release. • Share as...button saves output using a persistent data store based on room database – Will be implemented for subsequent play store release.

6.4.Requirement 4 – ~~Requirement removed due to further application analysis.~~

Title	Unit of measurement
Description	The user shall be able to specify the unit of measurement for output bandwidths
Mockups	Entry of MB, Entry of gb, Entry of bytes – Requirement no longer applicable.
Acceptance Tests	Validate rationale unit text
Test Results	Positive: MB, Negative foobar

7.

7. Design and Implementation

I utilized the new <https://claude.ai/> capability for render of diagrams without the need for Canvas, PlantUML, or other tools.

AI Prompt:

Design the basic architecture for an Android application that performs iperf3 tests against a remote iperf3 server

7.1. Layered Architecture

- The app is organized into four tiers from UI down to the native iperf3 binary.
- Unlike the AI provided Architecture, **I will not be utilizing an NDK/JNI Bridge** but instead will be executing the iperf3 binary and parsing its output.
- I decided to focus more effort on the video presentation.
- I will also be rearchitecting a solution to utilize JNI bindings and the Android Native Development Kit (NDK)

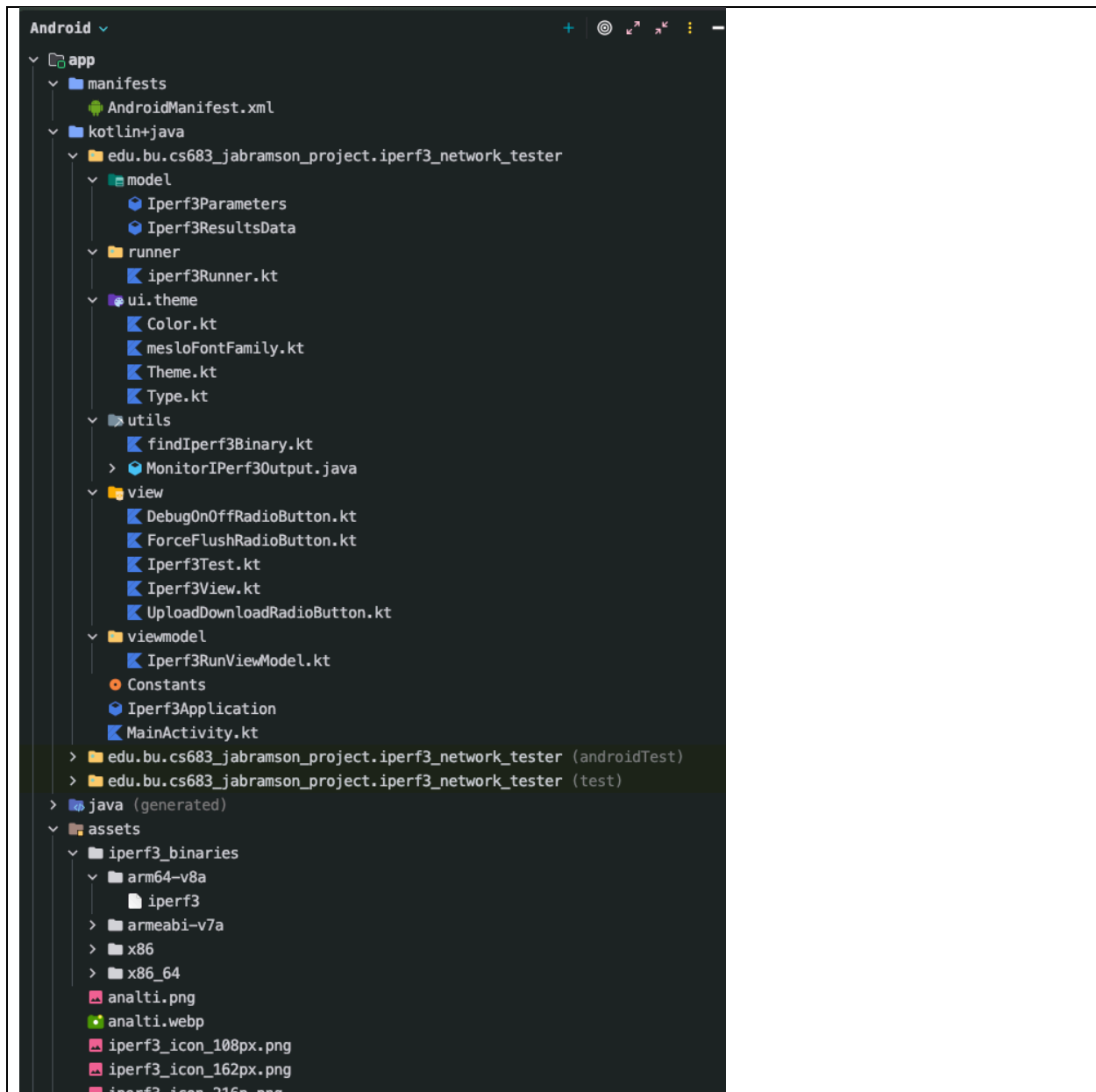
7.2. Test Execution Flow

Instead of utilizing the JNI bridge, I will be using the standard Kotlin/Java ProcessBuilder APIs to run the iperf3 binary in the background.

7.3. Full component breakdown organized by prose

I will be rewriting this section sometime after course completion.

2. Project Structure

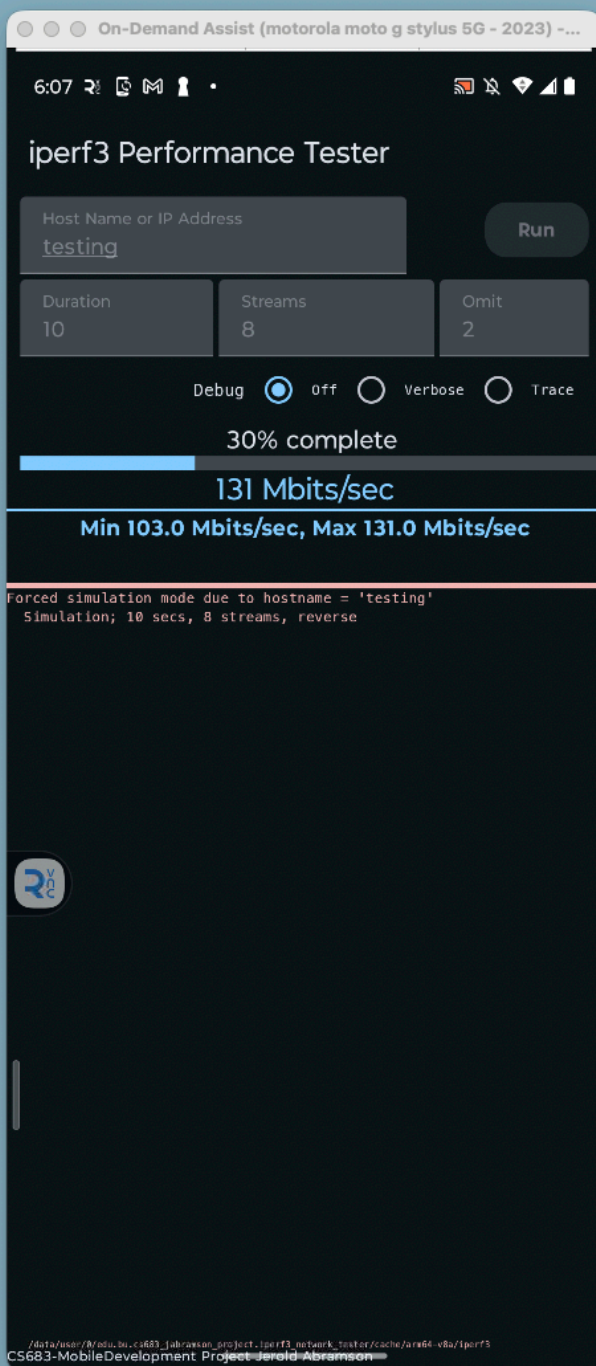


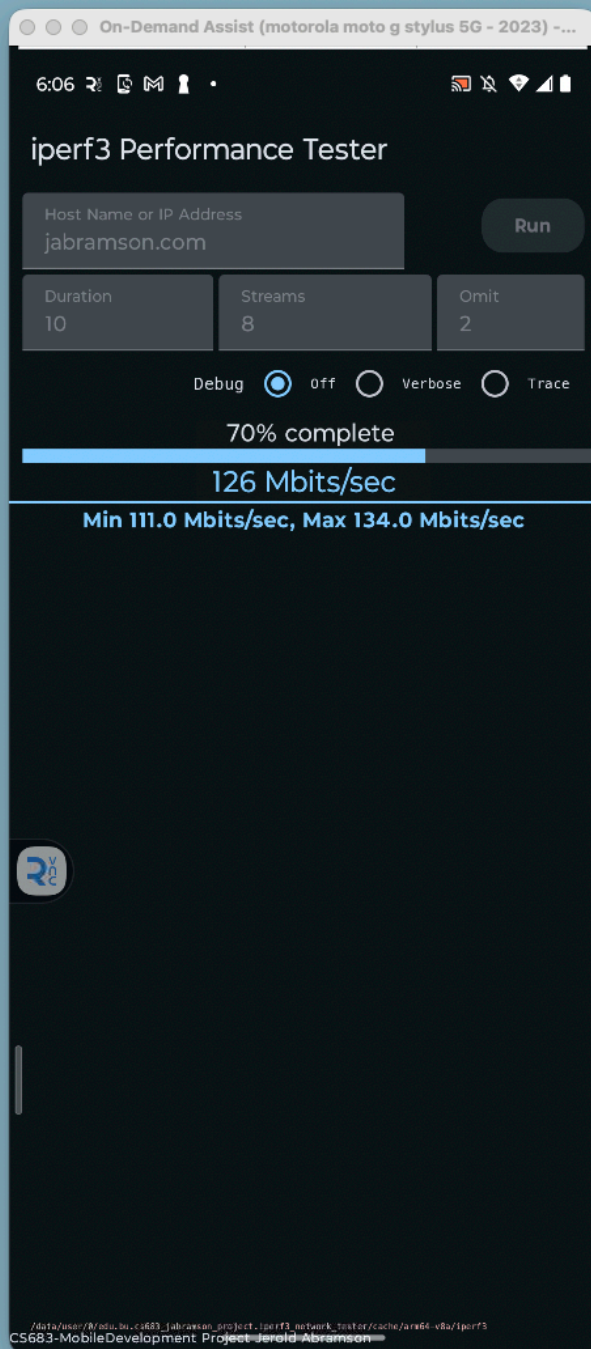
8.

9. Screen shots

9.1.Real Android Phone







9.2.Android Emulator

5:59



iperf3 Performance Tester

Host Name or IP Address

jabramson.com

Run

Duration

10

Streams

8

Omit

2

Debug



Off



Verbose



Trace

Launching iperf3 ...

iperf3 Performance Tester

Host Name or IP Address

jabramson.com

Run

Duration

10

Streams

8

Omit

2

Debug



Off



Verbose



Trace

19% complete

66.4 Mbits/sec

Min , Max

5:59   



iperf3 Performance Tester

Host Name or IP Address

jabramson.com

Run

Duration

10

Streams

8

Omit

2

Debug ☒ Off ☐ Verbose ☐ Trace

60% complete

186 Mbits/sec

Min 161.0 Mbits/sec, Max 186.0 Mbits/sec

5:59   



iperf3 Performance Tester

Host Name or IP Address

jabramson.com

Run

Duration

10

Streams

8

Omit

2

Debug



Off



Verbose



Trace

90% complete

176 Mbits/sec

Min 161.0 Mbits/sec, Max 186.0 Mbits/sec

5:59   



iperf3 Performance Tester

Host Name or IP Address

Run

Duration

Streams

Omit

Direction ☒ Download ☐ Upload

Debug ☒ Off ☐ Verbose ☐ Trace

Average: 179.0 Mbits/sec

Maximum: 188.0 Mbits/sec

Minimum: 161.0 Mbits/sec

5:58   



iperf3 Performance Tester

Host Name or IP Address
badhost.baddomain

Run

Duration
30

Streams
1

Omit
0

Direction ☒ Download ☐ Upload

Debug ☐ Off ☐ Verbose ☒ Trace

Return Code: 1

```
iperf3: error - unable to connect to server: No address  
associated with hostname
```

5:57   



iperf3 Performance Tester

Host Name or IP Address

jabramson.com

Run

Duration

30

Streams

1

Omit

0

Debug



Off



Verbose



Trace

6% complete

148 Mbits/sec

Min 148.0 Mbits/sec, Max 154.0 Mbits/sec

5:58   



iperf3 Performance Tester

Host Name or IP Address

1.1.1.1

Run

Duration

30

Streams

1

Omit

0

Direction ☒ Download ☐ Upload

Debug ☐ Off ☐ Verbose ☒ Trace

Return Code: 1

iperf3: error - unable to connect to server: Connection timed out

5:57



iperf3 Performance Tester

Host Name or IP Address

jabramson.com

Run

Duration

30

Streams

1

Omit

0

Debug



Off



Verbose



Trace

53% complete

165 Mbits/sec

Min 98.7 Mbits/sec, Max 165.0 Mbits/sec

Connecting to host jabramson.com, port 5201
Reverse mode, remote host jabramson.com is sending
Local Host/IP: 10.0.2.16
Remote Host/IP: 192.168.1.32
Remote Port: 5201

Running	0.00-1.00	154 Mbits/sec
Running	1.00-2.00	148 Mbits/sec
Running	2.00-3.00	129 Mbits/sec
Running	3.00-4.00	119 Mbits/sec
Running	4.00-5.00	100 Mbits/sec
Running	5.00-6.00	107 Mbits/sec
Running	6.00-7.00	111 Mbits/sec
Running	7.00-8.00	98.7 Mbits/sec
Running	8.00-9.00	115 Mbits/sec
Running	9.00-10.00	133 Mbits/sec
Running	10.00-11.00	138 Mbits/sec
Running	11.00-12.00	134 Mbits/sec
Running	12.00-13.00	135 Mbits/sec
Running	13.00-14.00	135 Mbits/sec
Running	14.00-15.00	160 Mbits/sec
Running	15.00-16.00	165 Mbits/sec

5:57   



iperf3 Performance Tester

Host Name or IP Address

jabramson.com

Run

Duration

30

Streams

1

Omit

0

Debug ☐ Off ☒ Verbose ☐ Trace

30% complete

115 Mbits/sec

Min 98.7 Mbits/sec, Max 154.0 Mbits/sec

```
Connecting to host jabramson.com, port 5201
Reverse mode, remote host jabramson.com is sending
Local Host/IP: 10.0.2.16
Remote Host/IP: 192.168.1.32
Remote Port: 5201
```

5:56   



iperf3 Performance Tester

Host Name or IP Address

Run

Duration

Streams

Omit

Direction ☐ Download ☒ Upload

Debug ☒ Off ☐ Verbose ☐ Trace

Average: 144.0 Mbits/sec

Maximum: 151.0 Mbits/sec

Minimum: 134.0 Mbits/sec

5:54   



iperf3 Performance Tester

Host Name or IP Address
jabramson.com

Run

Duration
10

Streams
8

Omit
2

Debug ☒ Off ☐ Verbose ☐ Trace

30% complete

149 Mbits/sec

Min 149.0 Mbits/sec, Max 149.0 Mbits/sec

10. Timeline

Iteration	Application Requirements • Essential • Desirable • Optional	Android Components and Features to be used
0.5	Redone to utilize Jetpack Compose	Jetpack Compose
1 DONE and thrown away	Essential: Stubbed version of progress bar	TBD (Networking)
COMPLETED 1	Essential: Application executes in background against a real host using the iperf3 executable	TextView
2 COMPLETED	As the iperf3 test is running in the background, provide a running total of the current bitrate	
2	After the test completes, provide a Min, Max, and Avg bitrate	
3 COMPLETED Test harness is in development	Ensure all possible network issues are handled gracefully	
2	Essential: Manual keyboard entry of test count	TextView
1		
2 or 3 COMPLETED EARLY Iteration 1	Essential: perform a single iperf3 test (count=1)	TBD (Consuming iperf3 native executable)
2 Handling iperf3 execution errors is adequate. Completed in iteration 4.	Desirable: Validate remote host	Networking
3 – DONE EARLY	Essential: Perform iperf3 to specified host with a count of 1	TBD
4 – DONE EARLY	Essential: iperf3 runs against specified host for specified count with specified unit	TBD

11. Current Status - Complete

- For the best experience use one of the simulators below.
- I have built in a simulation in case you have issues.
- I have also put together a small script that is executed prior to build.
 - This script changes the Android permissions to disable SELinux.
 - This will only work for certain simulators (include those shown below).
 - It will also work for root'ed Android devices.

If it is necessary to run the commands manually, these are the required steps:

```
$ adb -s ZD222CYWBG shell
genevn:/ $ su
genevn:/ # getenforce
Enforcing
genevn:/ # setenforce 0
genevn:/ # getenforce
Permissive
genevn:/ #
```

macOS: :Pixel 6 Android 16.0 ("Baklava") | arm64

Windows: Pixel 6 Android 16.0 ("Baklava") | x86_64



11.1. Fallback simulation

- If the test cannot execute an appropriate iperf3 executable, a simulation using actual iperf3 output is utilized instead.
- Additionall, if the hostname 'testing' is entered, the simulation is run.

12. AI Usage Log

All AI usage was tracked inside of the both the code and documentation directories as markdown files.

```
./Code/Iperf3NetworkTester/planning.md
./Code/Iperf3NetworkTester/app/src/main/structure.md
./Code/Iperf3NetworkTester/commandLineBuildApp.md
./Code/Iperf3NetworkTester/AGENTS.md
./Code/Iperf3NetworkTester/session-ses_2ba1.md
./Code/Iperf3NetworkTester/IPERF3_EXE.md
./Doc/02_answer.md
./Doc/iperf3Bundling/gradleTips.md
./Doc/iperf3Bundling/AGENTS.md
./Doc/android_icon_steps.md
./Doc/03_question.md
./Doc/01_question.md
./Doc/pre-deploy.md
./AGENTS.md
./CLAUDE.md
```

13. Future Work (Optional)

1. Rework application to utilize NDK and JNI bindings. I have already put together some prototypes of this solution.
2. Deploy to Android play store and begin marketing/sales.
3. Enhance video presentations and attempt to start a YouTube channel.

• Project Demo Links

Video Description	YouTube link
CS683 - Project - Part 1 of 4 - Jerold Abramson	https://youtu.be/duv60JoqVTs
CS683 - Project - Part 2 of 4 - Jerold Abramson	https://youtu.be/5vFkzoGIDoU
CS683 - Project - Part 3 of 4 - Jerold Abramson	https://youtu.be/2TtQ8dWUnhg
CS683 - Project - Part 4 of 4 - Jerold Abramson	https://youtu.be/iXRrtdUoofw

• References